

# Method comparison: original, production PVI, and our adaptation

| Aspect                             | Original 2D GCNM      | Production PVI                | Our PVI-GCNM                 |
|------------------------------------|-----------------------|-------------------------------|------------------------------|
| Problem type                       | absolute $\sigma$     | differential $\Delta\sigma$   | differential $\Delta\sigma$  |
| Physics per stage                  | recomputed            | one fixed step                | <b>recomputed</b>            |
| Jacobian                           | recomputed each stage | fixed at $\sigma_0 = 0.7$ S/m | <b>recomputed each stage</b> |
| <b>Prior filling the nullspace</b> | learned               | fixed smoothness $R$          | <b>learned</b>               |
| <b>Learned?</b>                    | yes                   | no                            | <b>yes</b>                   |
| Target                             | $\sigma^*$            | — none                        | clean $\Delta\sigma^*$       |
| Stages                             | 10                    | 1                             | 2                            |
| Coordinates                        | no                    | no                            | <b>yes:</b> $x, y, r$        |

**Takeaway.** Both PVI and ours are differential—that comes from PVI. Our change versus PVI is not the coordinate system; it is iterated, recomputed physics plus a learned prior replacing PVI’s single step and fixed smoothness prior.

**Thesis:** physics constrains the 32 measured directions; the learned prior regularizes the much larger unseen subspace.